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Section 5.5

Complex Eigenvalues

The Complex Conjugate

The complex conjugate of a complex number

$$z = a + bi$$

is

$$\bar{z} = a - bi$$

Example 1

Find the conjugates and the real and imaginary parts of each of the following.

$$z = 5 + 3i$$

$$z = 4 - 7i$$

$$\mathbf{x} = \begin{bmatrix} 5 - i \\ 2 \\ 3 + 5i \end{bmatrix}$$

Example 2

Find the eigenvalues and a basis for each eigenspace in $\mathbb{C}^2$.

$$A = \begin{bmatrix} 5 & -5 \\ 1 & 1 \end{bmatrix}$$

Complex Eigenvalues

Let A be an $n \times n$ matrix whose entries are real, and let λ be an eigenvalue of A with corresponding eigenvector $\mathbf{x}$. Then, $\bar{\lambda}$ is also an eigenvalue of A with corresponding eigenvector $\bar{\mathbf{x}}$.

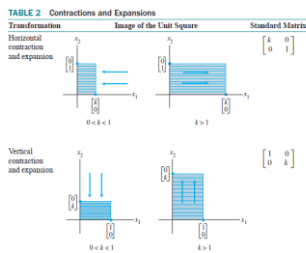
Example 3

Show that if a and b are real and not both zero, then the eigenvalues of $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ are $\lambda = a \pm bi$ with corresponding eigenvectors $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$.

Example 4

List the eigenvalues of $A = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$.

Recall: Common Transformations



Recall: Common Transformations

The standard matrix A of the transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which rotates each point in $\mathbb{R}^2$ about the origin through an angle φ (where positive angles correspond to counterclockwise rotation) is

$$A = \begin{bmatrix} \cos \varphi & -\sin \varphi \\ \sin \varphi & \cos \varphi \end{bmatrix}$$

Example 4 (continued)

The transformation $\mathbf{x} \mapsto A\mathbf{x}$ is the composition of a rotation and a scaling. Give the angle φ (where $-\pi < \varphi \leq \pi$) and give the scale factor r .

$$A = \begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix}$$

Fact

If $C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ where a and b are real and not both zero, then the eigenvalues of C are

$\lambda = a \pm bi$ with eigenvectors $\begin{bmatrix} 1 \\ -i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ i \end{bmatrix}$.

Also, if $r = |\lambda| = \sqrt{a^2 + b^2}$, then

$$C = r \begin{bmatrix} a/r & -b/r \\ b/r & a/r \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$$

so the transformation $\mathbf{x} \mapsto C\mathbf{x}$ is the composition of a rotation through φ and a scaling by $|\lambda|$.

Theorem

Let A be a real 2×2 matrix with a complex eigenvalue $\lambda = a - bi$ (with $b \neq 0$) and an associated eigenvector $\mathbf{v}$ in $\mathbb{C}^2$. Then,

$$A = PCP^{-1}$$

where

$$P = [\operatorname{Re} \mathbf{v} \quad \operatorname{Im} \mathbf{v}]$$

and

$$C = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$$

Example 5

Find an invertible matrix P and a matrix C of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ such that the given matrix has the form $A = PCP^{-1}$.

$$A = \begin{bmatrix} 5 & -2 \\ 1 & 3 \end{bmatrix}$$
